# Guojie Ruan

## BASIC INFORMATION

Add.: Columbia, Missouri, USA
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Academic website: https://ruanguojie.github.io/

## RESEARCH INTERESTS

Precision nitrogen management
Agroecosystem remote sensing
Cropland soil health
Plant phenotyping

## EDUCATION

**University of Missouri,** Columbia, USA — 2024-Now
Precision and Automated Agriculture Lab
PhD student in Plant, Insect and Microbial Sciences

**Nanjing Agricultural University**, Nanjing, China — 2020-2023
National Engineering and Technology Center for Information Agriculture
M.Sc., Crop Cultivation and Farming, *"Double First-class" disciplines*
Thesis: Winter wheat smart nitrogen recommendation algorithms based on multi-source data fusion

**South China Agricultural University**, Guangzhou, China — 2016-2020
B.Sc., Agronomy (Ding Ying Innovation Program), *"Double First-class" disciplines*
Overall GPA: 4.04/5.0

## RESEARCH EXPERIENCE

**Graduate Research Assistant, Division of Plant Science and Technology, Mizzou**

- Advisor: Jianfeng Zhou

**Participating in Jiangsu Provincial Key R&D Program, NJAU**

- Project name: Precision Design and Engineering Implementation of Intelligent Production Management Prescriptions for Rice and Wheat.
- Advisor: Professor Qiang Cao
- Developed data-driven based winter wheat topdressing nitrogen management framework with proximal sensing, meteorological, and soil information.
- Writing project applications and research papers.
- Led a 'College Students' Innovative Entrepreneurial Training Plan Program' undergraduate team and secured national-level funding.

**Completing the Graduation Project of Undergraduate, SCAU**

- Project name: Comparative study on Seed Storability of Chinese Soybean Landrace Population.
- Advisor: Professor Cunyi Yang
- Conducted controlled deterioration treat using 364 soybean varieties.
- Collected germination index of seed vigor and provided data for GWAS.

**JOURNAL PUBLICATIONS**

Publication Metrics: **166 citations** in total, accessed on 28 August 2026

1. **Ruan, G.**, Li, X., Yuan, F., Cammarano, D., Ata-UI-Karim, S. T., Liu, X., Tian, Y., Zhu, Y., Cao, W., Cao, Q.*, 2022. Improving wheat yield prediction integrating proximal sensing and weather data with machine learning. *Computers and Electronics in Agriculture*, 195, 106852. Doi: 10.1016/j.compag.2022.106852 (IF=10.3, Q1)
2. **Ruan, G.**, Schmidhalter, U., Yuan, F., Cammarano, D., Liu, X., Tian, Y., Zhu, Y., Cao, W., Cao, Q.*, 2023. Exploring the transferability of wheat nitrogen status estimation with multisource data and Evolutionary Algorithm-Deep Learning (EA-DL) framework. *European Journal of Agronomy*, 143, 126727.
Doi: 10.1016/j.eja.2022.126727 (IF=6.7, Q1)
3. **Ruan, G.**, Cammarano, D., Ata-UI-Karim, S. T., Liu, X., Tian, Y., Zhu, Y., Cao, W., Cao, Q.*, 2024. Investigating data-driven approaches to optimize nitrogen recommendations for winter wheat. *Computers and Electronics in Agriculture*, 220, 108857.
Doi: 10.1016/j.compag.2024.108857 (IF=10.3, Q1)
4. Song H.#, Zhuang T.#, Li X., **Ruan G.**, Schepers J., Wang D., Liu, X., Tian, Y., Zhu, Y., Cao, W., Cao, Q.*, 2025. Enhancing winter wheat growth indicator prediction with multi-task learning and multi-source data. *European Journal of Agronomy*, 168, 127629. Doi: 10.1016/j.eja.2025.127629 (IF=6.7, Q1)
5. **Ruan, G.**, Tian, F., Reinbott, T., Aloysius, N., Hunt, S., & Zhou, J. (2026). Developing high-resolution digital soil health maps for soybean field using multiple-source sensor data. Smart Agricultural Technology, 102408. Doi: 10.1016/j.atech.2026.102408 (IF=7.1)

**CONFERENCE PRESENTATIONS**

1. Toward High-Resolution Digital Soil Health Mapping Using Sensor Data. American Geophysical Union (AGU) 2025 Annual Meeting Poster Presentation, New Orleans, LA.
2. Spatiotemporal Mapping of Corn Leaf Nutrients: Integrating UAV-based RGB and Hyperspectral Features. American Society of Agricultural & Biological Engineers (ASABE) 2026 Annual International Meeting Oral Presentation, Indianapolis, IN.
3. MU Digital Agricultural Symposium 2025&2026, Poster Presentation.

**OUTREACH**

Mizzou College of Agriculture, Food and Natural Resources Showcase, 2025
Mizzou Tomato Festival, 2025&2026
Founder & Editor, FieldStar. WeChat science-communication platform on smart agriculture and Earth system science, 2020~Present. 13000+ followers.

**AWARDS**

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| National Scholarships, Ministry of Education, China | 2022 |
| Outstanding Graduate Student of NJAU | 2023 |
| Outstanding Graduate Student of SCAU | 2020 |
| First-Class Academic Scholarship of NJAU | 2021 |
| Third-Class Academic Scholarship (Undergraduate) of SCAU | 2018 |
| Excellent Master Dissertation Award of NJAU | 2023 |
| Excellent Master Dissertation Award of Jiangsu Province | 2024 |

**SKILLS**

Language: Mandarin (Native), English, Cantonese (Native)
Computer Programming: Python, R
Software & Tools: ENVI, ArcGIS Pro, QGIS, SPSS Statistics,
Agisoft, Headwall, Google Earth Engine
Experimental Skills: Drone operation, Soil sensor